St. Joseph's Convent Sr.Sec. School

Jalgaon

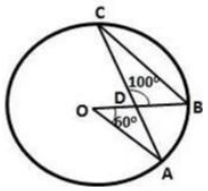
Mid Term Exam

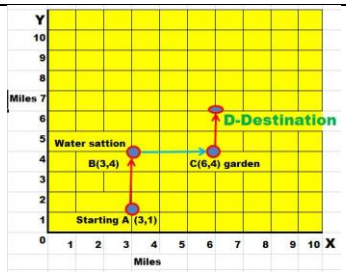
CLASS 09 - MATHS
Time Allowed: 2 hours and 59 minutes Maximum Marks : 60

	Section A	
1	The point P lies on the line y = 2x − 3 and its abscissa is 7. The ordinate of P is: a) 4 b) 11 c) 17 d) 7	**[1]**
2	If the point (a − 2, 3a) lies on the x-axis, then the value of a is: a) 0 b) 2 c) −2 d) 3	**[1]**
3	A cab charges a fixed fee of ₹55 and ₹14 per kilometre. If the distance travelled is 15 km, the total fare is: a) ₹210 b) ₹251 c) ₹265 d) ₹279	**[1]**
4	Evaluate: 125^(1/3) − 81^(1/4) + 32^(1/5) a) 4 b) 5 c) 6 d) 10	**[1]**
5	Express 3/11 as a decimal: a) 0.333… (3 repeating) b) 0.2727… (27 repeating) c) 0.27 d) 0.311	**[1]**
6	Simplify: $(0.8^3 + 0.3^3) \div (0.8^2 - 0.8 \times 0.3 + 0.3^2)$ a) 0.5 b) 1.1 c) 0.8 d) 0.3	**[1]**

7	Expand: $(3x - 2)(2x + 1)$ a) $6x^2 - x - 2$ b) $6x^2 + x - 2$ c) $6x^2 - 7x - 2$ d) $6x^2 + 7x + 2$	**[1]**
8	ABCD is a cyclic quadrilateral. If $\angle B = 3x + 10°$ and $\angle D = 5x - 6°$, then $\angle D$ equals: a) 96° b) 100° c) 104° d) 108°	**[1]**
9	How many circles can pass through three distinct collinear points? a) 0 b) 1 c) 2 d) Infinitely many	**[1]**
10	The diagonals of a rhombus are in the ratio 5:12 and its side is 26 cm. The longer diagonal is: a) 24 cm b) 26 cm c) 48 cm d) 50 cm	**[1]**
11	A triangle has base 12 cm and height 7 cm. A second triangle has the same area and height 14 cm. Its base is: a) 3 cm b) 6 cm c) 12 cm d) 21 cm	**[1]**
12	A die is thrown once. The probability of getting a number that is neither prime nor greater than 4 is: a) 1/6 b) 1/3 c) 1/2 d) 2/3	**[1]**
13	Assertion (A): The degree of the polynomial $7x^4 - 3x^2 + 9$ is 9. Reason (R): The degree of a polynomial is the greatest exponent of the variable with a non-zero coefficient. a) Both A and R are true, and R correctly explains A b) Both A and R are true, but R does not explain A	**[1]**

	c) A is true, but R is false d) A is false, but R is true	
14	Assertion (A): Two distinct lines can have two common points. Reason (R): Through two distinct points, exactly one line can be drawn. a) Both A and R are true, and R correctly explains A b) Both A and R are true, but R does not explain A c) A is true, but R is false d) A is false, but R is true	**[1]**
15	A card is chosen at random from cards numbered 1 to 20. The probability that the number is divisible by neither 3 nor 5 is: a) 1/4 b) 2/5 c) 9/20 d) 11/20	[1]
16	If $(4x + 10)°$ and $(2x + 20)°$ are complementary angles, then x is: a) 5 b) 10 c) 15 d) 20	[1]
17	The pair of equations represented by two parallel lines has: a) no solution b) exactly one solution c) infinitely many solutions d) exactly two solutions	[1]
18	If $p(x)=2x^3-5x^2+x+4$, then $p(1)$ is: a) −4 b) −2 c) 2 d) 4	[1]
	Section B	
19	Rewrite $5x - 2y - 11 = 0$ in the form $y = mx + c$. State m and c, and verify whether (2, 3) is a solution.	**[1]**
20	Factorise completely: $8x^3 - 125y^3$.	**[2]**
21	If $x - y = 5$ and $xy = 14$, find $x^3 - y^3$ without finding x and y separately.	**[2]**
22	A box contains 6 red, 5 blue and 9 yellow counters. One counter is selected at random. Find the probability that it is (i) blue, and (ii) not red.	**[1]**

23	The point (2, −5) lies on the line 3x − 2y = k. Find k and then write two other integer-coordinate points that satisfy the same equation.	**[3]**
24	Rationalise the denominator and simplify: $(\sqrt{13} + \sqrt{5})/(\sqrt{13} - \sqrt{5})$. Express the result in the form $a + b\sqrt{65}$.	**[3]**
25	Using the Factor Theorem, factorise $x^3 + 4x^2 - x - 4$ completely.	**[3]**
26	In the following figure, Ois centre of the circle,$\angle AOB = 60°$ and $\angle BDC = 100°$. Find $\angle OAC$. 	**[3]**
27	The sides of a triangular field are in the ratio 13:14:15 and its area is 5376 m^2. Find its perimeter.	**[2]**
28	A delivery service charges ₹18 for the first kilometre and ₹7 for each additional kilometre or part thereof. Let x be the total distance in km and y the fare for a whole-number distance. (i) Write x as a linear expression in y. (ii) Find the distance travelled for a fare of ₹95. (iii) A customer travels 9 km; how much do they pay?	**[3]**
29	In a circle with centre O, AB is a diameter and C is a point on the circle. If ∠BAC = 63°, find ∠ABC and ∠AOC, giving reasons. If a chord CD is perpendicular to AB, show that AB bisects CD and find the angle subtended by CD at the centre.	**[3]**
30	The sides of a triangle are 25 cm, 39 cm and 56 cm. Use Heron's formula to find its area. Hence calculate the altitude to the side of length 56 cm.	**[3]**
31	Using algebraic identities, express $3(2x - 3y)(3y - 4z)(4z - 2x)$ as a sum of three cubes. State the identity used.	[3]
32	Two parallel lines are cut by a transversal. A pair of alternate interior angles are $(3x + 15)°$ and $(5x + 5)°$. Find x and all the angles formed.	[3]
33	A triangular plot has sides 10 m, 17 m and 21 m. Find its area using Heron's formula and then find the cost of fencing its boundary at ₹12 per m.	[3]
	Section D	
34	**Read the following text carefully and answer the questions that follow:**	**[3]**



Arun has completed an **8 miles** walk and now returns along the same course, which was plotted on a square coordinate grid. The return journey starts at the destination D (6, 6).

From the destination D, the walkway goes down to the garden at C (6, 4). From the garden, the walkway turns left and reaches the water station at B (3, 4). From the water station, the walkway goes down and finally Arun reaches the starting point A to complete the return journey.

1. How far is the garden C from destination D? (1)
2. What is the abscissa of starting point A? (1)
3. What is the ordinate of starting point A? (2)

   OR

   What are the coordinates of starting point A? (2)

35 A company selected 4000 households at random and surveyed them to find out a relationship between income level and the number of television sets in a home. The information so obtained is listed in the following table: **[2]**

Monthly income	**Number of Television/household**			
	0	**1**	**2**	**Above 2**
10000	20	80	10	0
10000-14999	10	240	60	0
15000-19999	0	380	120	30
20000-24999	0	520	370	80
25000 and above	0	1100	760	220

find the probability:

1. that a household is NOT one earning₹ 10000 - ₹ 14999 per year and having exactly one television.

	2. that a household is NOT one earning₹ 25000 and more per year and owning 2 televisions. 3. of a household having at least one television. Answer any **TWO** of the above.	